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Chapter 5

Playing with Numbers

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Exercise 5(A)



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1. 5
2. 10
3. 11
4. none of these

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Answer

According to the property, when $ab + ba$ is divided by $a + b$, the quotient = 11.

So, when $68 + 86$ is divided by 14 ($6 + 8 = 14$), the quotient = 11.

Hence, option 3 is the correct option.

Question 1(ii)

The quotient when $75 - 57$ is divided by 9 is :

1. 4

2. 2

3. 9

4. none of these

Answer

According to the property, when $ab - ba$ is divided by 9, the quotient = $a - b$.

So, when $75 - 57$ is divided by 9, the quotient = $7 - 5 = 2$.

Hence, option 2 is the correct option.

Question 1(iii)

For

$$\begin{array}{r} 1 \text{ A} \\ + \text{ A } 3 \\ \hline 1 \text{ 0 } 1 \end{array}$$

the value of digit A is :

1. 18
2. 6
3. 28
4. none of these

Answer

$$\begin{array}{r} 1 A \\ + A 3 \\ \hline 1 0 \end{array}$$

$A + 3$ must give a number whose ones digit is 1.

$$\Rightarrow A + 3 = 1, A + 3 = 11, A + 3 = 21, \dots$$

$$\Rightarrow A = 1 - 3, A = 11 - 3, A = 21 - 3, \dots$$

$$\Rightarrow A = -2, A = 8, A = 18, \dots$$

$$\Rightarrow A = 8, \text{ as } A \text{ is a digit.}$$

$$\begin{array}{r} 1 8 \\ + 8 3 \\ \hline 1 0 \end{array}$$

Hence, option 4 is the correct option.

Question 1(iv)

The quotient, when difference of the number 53 and the number obtained on reversing its digits is divided by 2 :

1. 8

2. 9

3. 3

4. 18

Answer

According to the property, when $ab - ba$ is divided by $a - b$, the quotient = 9.

So, when $53 - 35$ is divided by 2 ($5 - 3 = 2$), the quotient = 9

Hence, option 2 is the correct option.

Question 1(v)

For

$$\begin{array}{r} 1 2 A \\ - A 9 \\ \hline 4 8 \\ \hline \end{array}$$

the value of digit A is :

1. 6

2. 7

3. -1

4. none of these

Answer

$$\begin{array}{r}
 1 \quad 2 \quad A \\
 - \quad \quad A \quad 9 \\
 \hline
 \quad \quad 4 \quad 8 \\
 \hline
 \end{array}$$

A - 9 must give a number whose ones digit is 8.

$$\Rightarrow A - 9 = 8, A - 9 = 18, A - 9 = 28, \dots\dots\dots$$

$$\Rightarrow A = 8 + 9, A = 18 + 9, A = 28 + 9, \dots\dots\dots$$

$$\Rightarrow A = 17, A = 27, A = 37, \dots\dots\dots$$

$$\Rightarrow A = 7, \text{ as } A \text{ is a digit.}$$

Hence, option 2 is the correct option.

Question 2(i)

Find the quotient, when the sum of 94 and 49 is divided by :

(a) 11

(b) 13

Answer

(a) According to the property, when $ab + ba$ is divided by 11, the quotient = $a + b$.

So, when $94 + 49$ is divided by 11, **the quotient = $9 + 4 = 13$**

(b) According to the property, when $ab + ba$ is divided by $a + b$, the quotient = 11.

So, when $94 + 49$ is divided by 13 ($9 + 4 = 13$), **the quotient = 11.**

Question 2(ii)

Find the quotient when the difference between 94 and 49 is divided by :

(a) 5

(b) 9

Answer

(a) According to the property, when $ab - ba$ is divided by $a - b$, the quotient = 9.

So, when $94 - 49$ is divided by 5 ($9 - 4 = 5$), the quotient = 9

(b) According to the property, when $ab - ba$ is divided by 9, the quotient = $a - b$.

So, when $94 - 49$ is divided by 9, the quotient = $(9 - 4) = 5$.

Question 3

If $a = b$, show that $abc = bac$.

Answer

$$abc = bac$$

$$\Rightarrow a \times b \times c = b \times a \times c$$

$$\Rightarrow a \times a \times c = a \times a \times c$$

$$\Rightarrow a^2 \times c = a^2 \times c$$

$$\Rightarrow \cancel{a^2} \times c = \cancel{a^2} \times c$$

$$\Rightarrow c = c$$

$\therefore$ If $a = b$, then $abc = bac$

Question 4

Write the quotient when the difference between 694 and number obtained on interchanging its ones and hundreds digits is divided by :

(i) 33

(ii) 99

(iii) 2

Answer

Lets take the 3-digit number as abc

$\therefore$ Number obtained on interchanging its ones and hundreds digits will be cba

$$abc - cba$$

$$= (100a + 10b + c) - (100c + 10b + a)$$

$$= 100a + 10b + c - 100c - 10b - a$$

$$= (100a - a) + (10b - 10b) + (c - 100c)$$

$$= (99a) + (-99c)$$

$$= 99(a - c)$$

$$= 99(a - c)$$

Here $a = 6$, $b = 9$ and $c = 4$

(i) On dividing by 33:

$$\frac{99(a - c)}{33}$$

$$= 3(a - c)$$

Substituting values of a and c we get,

$$3(6 - 4) = 3 \times 2 = 6$$

$\therefore$ When the difference between 694 and number obtained on interchanging its ones and hundreds digits is divided by 33, its quotient will be 6.

(ii) On dividing by 99:

$$\frac{99(a - c)}{99}$$

$$= (a - c)$$

Substituting values of a and c we get,

$$6 - 4 = 2$$

$\therefore$ When the difference between 694 and number obtained on interchanging its ones and hundreds digits is divided by 99, its quotient will be 2.

(iii) On dividing by 2:

$$\frac{99(a - c)}{2}$$

Substituting values of a and c we get,

$$\begin{aligned} & \frac{99(6 - 4)}{2} \\ &= \frac{99 \times 2}{2} \\ &= 99 \end{aligned}$$

$\therefore$ When the difference between 694 and number obtained on interchanging its ones and hundreds digits is divided by 2, its quotient will be 99.

Question 5

If the sum of the number 842 and two other numbers obtained by arranging the digits in cyclic order is divided by :

- (i) 111
- (ii) 14
- (iii) 37

Answer

(i) According to the property 3, when $abc + bca + cab$ is divided by 111, the quotient = $a + b + c$.

Here the number is 842 and when it is arranged in cyclic order it becomes 428 and 284.

When $842 + 428 + 284$ is divided by 111, the quotient = $8 + 4 + 2 = 14$.

(ii) According to the property 3, when $abc + bca + cab$ is divided by $a + b + c$, the quotient = 111 .

Here the number is 842 and when it arrange in cyclic order it becomes 428 and 284.

When $842 + 428 + 284$ is divided by $8 + 4 + 2$ (14), the quotient = 111.

(iii) Here the number is 842 and when it arrange in cyclic order it becomes 428 and 284.

$$\begin{aligned} & \frac{842 + 428 + 284}{37} \\ &= \frac{1554}{37} \\ &= 42 \end{aligned}$$

When $842 + 428 + 284$ is divided by 37, the quotient = 42.

Question 6

If $a * b = a + 2b + 5$, then find the value of $4 * 5$.

Answer

$$\text{If } a * b = a + 2b + 5$$

Putting $a = 4$ and $b = 5$, we get

$$4 * 5 = 4 + 2 \times 5 + 5$$

$$\Rightarrow 4 * 5 = 4 + 10 + 5$$

$$\Rightarrow 4 * 5 = 19$$

Hence, $4 * 5 = 19$

Question 7

If $a * b = 2a - b - 2$, find the value of $8 * 7$.

Answer

If $a * b = 2a - b - 2$

Putting $a = 8$ and $b = 7$, we get

$$8 * 7 = 2 \times 8 - 7 - 2$$

$$\Rightarrow 8 * 7 = 16 - 7 - 2$$

$$\Rightarrow 8 * 7 = 7$$

Hence, $8 * 7 = 7$

Question 8(a)

In each of the following cases, find the least value/values of letters used in place of digits :

$$\begin{array}{r} 3 \quad A \\ + \quad 2 \quad 5 \\ \hline B \quad 2 \end{array}$$

Answer

$$\begin{array}{r} 3 \quad A \\ + \quad 2 \quad 5 \\ \hline B \quad 2 \end{array}$$

Clearly, $A + 5$ is a number whose ones digit is 2.

$$\Rightarrow A + 5 = 2, A + 5 = 12, A + 5 = 22; \text{ and so on.}$$

$\Rightarrow A = 2 - 5, A = 12 - 5, A = 22 - 5$; and so on.

$\Rightarrow A = -3, A = 7, A = 17$; and so on.

As A is a digit, so $A = 7$.

$$\begin{array}{r} \overset{1}{3} 7 \\ + 2 5 \\ \hline B 2 \end{array}$$

Now, finding the value of B.

$$1 + 3 + 2 = B$$

$$6 = B$$

Since, A is a digit. $\therefore A = 7, B = 6$.

Question 8(b)

In each of the following cases, find the least value/values of letters used in place of digits :

$$\begin{array}{r} 9 8 \\ + 4 A \\ \hline C B 3 \end{array}$$

Answer

$$\begin{array}{r} 9 8 \\ + 4 A \\ \hline C B 3 \end{array}$$

Firstly, we will find the value of A.

Clearly, $8 + A$ is a number whose ones digit is 3.

$\Rightarrow 8 + A = 3, 8 + A = 13, 8 + A = 23$; and so on.

$\Rightarrow A = 3 - 8, A = 13 - 8, A = 23 - 8$; and so on.

$\Rightarrow A = -5, A = 5, A = 15$; and so on.

Since, A is a digit. $\therefore A = 5$

$$\begin{array}{r} \overset{1}{9} 8 \\ + 4 5 \\ \hline C B 3 \end{array}$$

Secondly, we will find the value of B

$$1 + 9 + 4 = B$$

$$14 = B$$

Since, B is a digit. $\therefore B = 4$ and hence, $C = 1$

$$\begin{array}{r} 9 8 \\ + 4 5 \\ \hline 1 4 3 \end{array} \quad \mathbf{A = 5, B = 4 \text{ and } C = 1}$$

Question 8(c)

In each of the following cases, find the least value/values of letters used in place of digits :

$$\begin{array}{r} A 1 \\ + 1 B \\ \hline B 0 \end{array}$$

Answer

$$\begin{array}{r}
 A \quad 1 \\
 + \quad 1 \quad B \\
 \hline
 B \quad 0
 \end{array}$$

Firstly we will find the value of B.

Clearly, $1 + B$ is a number whose ones digit is 0.

$\Rightarrow 1 + B = 0, 1 + B = 10, 1 + B = 20$; and so on.

$\Rightarrow B = 0 - 1, B = 10 - 1, B = 20 - 1$; and so on.

$\Rightarrow B = -1, B = 9, B = 19$; and so on.

Since, B is a digit, $\therefore B = 9$.

Secondly we will find the value of A.

$$\begin{array}{r}
 \overset{1}{A} \quad 1 \\
 + \quad 1 \quad 9 \\
 \hline
 9 \quad 0
 \end{array}$$

Clearly, 1 is carried over on tens place. $A + 1 + 1 = 9$.

$\Rightarrow A + 2 = 9, A + 2 = 19, A + 2 = 29$; and so on.

$\Rightarrow A = 9 - 2, A = 19 - 2, A = 29 - 2$; and so on.

$\Rightarrow A = 7, B = 17, B = 27$; and so on.

Since, A is a digit. $\therefore A = 7$.

$$\begin{array}{r}
 7 \quad 1 \\
 + \quad 1 \quad 9 \\
 \hline
 9 \quad 0
 \end{array}
 \quad \mathbf{A = 7 \text{ and } B = 9}$$

Question 8(d)

In each of the following cases, find the least value/values of letters used in place of digits :

$$\begin{array}{r} 2 \quad A \quad B \\ + \quad A \quad B \quad 1 \\ \hline B \quad 1 \quad 8 \end{array}$$

Answer

$$\begin{array}{r} 2 \quad A \quad B \\ + \quad A \quad B \quad 1 \\ \hline B \quad 1 \quad 8 \end{array}$$

Firstly, we will find the value of B.

Clearly, $B + 1$ is a number whose ones digit is 8.

$\Rightarrow B + 1 = 8, B + 1 = 18, B + 1 = 28$; and so on.

$\Rightarrow B = 8 - 1, B = 18 - 1, B = 28 - 1$; and so on.

$\Rightarrow B = 7, B = 17, B = 27$; and so on.

Since, B is a digit. $\therefore B = 7$.

$$\begin{array}{r} 2 \quad A \quad 7 \\ + \quad A \quad 7 \quad 1 \\ \hline 7 \quad 1 \quad 8 \end{array}$$

Secondly, we will find the value of A.

Clearly, $A + 7$ is a number whose ones digit is 1.

$\Rightarrow A + 7 = 1, A + 7 = 11, A + 7 = 21$; and so on.

$\Rightarrow A = 1 - 7, A = 11 - 7, A = 21 - 7$; and so on.

$\Rightarrow A = -6, A = 4, A = 14$; and so on.

Since, A is a digit. $\therefore A = 4$.

$$\begin{array}{r} 2 \quad 4 \quad 7 \\ + \quad 4 \quad 7 \quad 1 \\ \hline 7 \quad 1 \quad 8 \end{array}$$

A = 4 and B = 7.

Question 8(e)

In each of the following cases, find the least value/values of letters used in place of digits :

$$\begin{array}{r} 1 \quad 2 \quad A \\ + \quad 6 \quad A \quad B \\ \hline A \quad 0 \quad 9 \end{array}$$

Answer

$$\begin{array}{r} 1 \quad 2 \quad A \\ + \quad 6 \quad A \quad B \\ \hline A \quad 0 \quad 9 \end{array}$$

Firstly, we will find the value of A.

Clearly, $2 + A$ is a number whose ones digit is 0.

$\Rightarrow 2 + A = 0, 2 + A = 10, 2 + A = 20$; and so on.

$\Rightarrow A = 0 - 2, A = 10 - 2, A = 20 - 2$; and so on.

$\Rightarrow A = -2, A = 8, A = 18$; and so on.

Since, A is a digit. $\therefore A = 8$.

$$\begin{array}{r} \overset{1}{1} 2 8 \\ + 6 8 B \\ \hline 8 0 9 \end{array}$$

Secondly, we will find the value of B.

Clearly, $8 + B$ is a number whose ones digit is 9.

$\Rightarrow 8 + B = 9, 8 + B = 19, 8 + B = 29$; and so on.

$\Rightarrow B = 9 - 8, B = 19 - 8, B = 29 - 8$; and so on.

$\Rightarrow B = 1, B = 11, B = 21$; and so on.

Since, B is a digit. $\therefore B = 1$.

$$\begin{array}{r} \overset{1}{1} 2 8 \\ + 6 8 1 \\ \hline 8 0 9 \end{array}$$

A = 8 and B = 1.

Question 8(f)

In each of the following cases, find the least value/values of letters used in place of digits :

$$\begin{array}{r} 1 A \\ \times A \\ \hline 9 A \end{array}$$

Answer

$$\begin{array}{r} 1 \quad A \\ \times \quad A \\ \hline 9 \quad A \\ \hline \end{array}$$

Clearly, we will find the value of A.

$$A \times A = A$$

$$\Rightarrow A^2 = A$$

Find a number whose square gives the same number on unit place.

$$1 \times 1 = 1$$

$$5 \times 5 = 25$$

$$6 \times 6 = 36$$

Lets take A = 1

$$\begin{array}{r} 1 \quad 1 \\ \times \quad 1 \\ \hline 1 \quad 1 \\ \hline \end{array}$$

Hence, 1 does not satisfy the equation.

Lets take A = 5

$$\begin{array}{r} 1^2 \quad 5 \\ \times \quad 5 \\ \hline 7 \quad 5 \\ \hline \end{array}$$

5 also does not satisfy the multiplication.

Lets take $A = 6$

$$\begin{array}{r} \overset{3}{1} \ A \\ \times \ A \\ \hline 9 \ A \end{array}$$

Hence, $A = 6$.

Question 8(g)

In each of the following cases, find the least value/values of letters used in place of digits :

$$\begin{array}{r} \ A \ B \\ \times \ 6 \\ \hline B \ B \ B \end{array}$$

Answer

$$\begin{array}{r} \ A \ B \\ \times \ 6 \\ \hline B \ B \ B \end{array}$$

Firstly, we will find the value of B.

Hence, $B \times 6 = B$

A number multiplied with 6 gives same number in its unit place.

$$2 \times 6 = 12$$

$$4 \times 6 = 24$$

$$6 \times 6 = 36$$

$$8 \times 6 = 48$$

Hence, B = 2 or 4 or 6 or 8

Now find the value of A

Lets take B = 2

Then

$$\begin{array}{r} \overset{1}{A} \quad 2 \\ \times \quad 6 \\ \hline 2 \quad 2 \quad 2 \end{array}$$

1 carried on A means $A \times 6 + 1 = 22$. Since no value of A satisfy the equation.

Hence $B \neq 2$.

Lets take B = 4

Then

$$\begin{array}{r} \overset{2}{A} \quad 4 \\ \times \quad 6 \\ \hline 4 \quad 4 \quad 4 \end{array}$$

2 carried on A means $A \times 6 + 2 = 44 \Rightarrow A \times 6 = 42 \Rightarrow A = 7$.

Lets take B = 6

Then

$$\begin{array}{r}
 ^3 \\
 \text{A} \quad 6 \\
 \times \quad 6 \\
 \hline
 6 \quad 6 \quad 6
 \end{array}$$

3 carried on A means $A \times 6 + 3 = 66$. Since no value of A satisfy the equation.

Hence $B \neq 6$.

Lets take $B = 8$

Then

$$\begin{array}{r}
 ^4 \\
 \text{A} \quad 8 \\
 \times \quad 6 \\
 \hline
 8 \quad 8 \quad 8
 \end{array}$$

4 carried on A means $A \times 6 + 4 = 88$. Hence $A = 14$. But A is single digit number.

So, $B \neq 8$

$$\begin{array}{r}
 ^2 \\
 \text{A} \quad 4 \\
 \times \quad 6 \\
 \hline
 4 \quad 4 \quad 4
 \end{array}$$

Hence, $A = 7$ and $B = 4$

Question 8(h)

In each of the following cases, find the least value/values of letters used in place of digits :

$$\begin{array}{r}
 \text{A} \quad \text{B} \\
 \times \quad 3 \\
 \hline
 \text{C} \quad \text{A} \quad \text{B}
 \end{array}$$

Answer

$$\begin{array}{r} \text{A} \text{ B} \\ \times \quad 3 \\ \hline \text{C} \text{ A} \text{ B} \end{array}$$

Firstly, we will find the value of B.

$$B \times 3 = B$$

$$0 \times 3 = 0$$

$$5 \times 3 = 15$$

So, 0 and 5 are possible values of B.

Now finding the value of A.

Lets take B = 0 then A x 3 = A

$$\begin{array}{r} \text{A} \text{ 0} \\ \times \quad 3 \\ \hline \text{C} \text{ A} \text{ 0} \end{array}$$

$$0 \times 3 = 0$$

$$5 \times 3 = 15$$

As we need 2-digit number in answer. So A = 5 and C = 1.

Lets take B = 5

$$\begin{array}{r} 1 \\ A 5 \\ \times 3 \\ \hline C A 5 \end{array}$$

$$A \times 3 + 1 = A$$

$$\Rightarrow A \times 3 = A - 1$$

There is no value for A which satisfies the equation. Hence $B \neq 5$

$$\begin{array}{r} 5 0 \\ \times 3 \\ \hline 1 5 0 \end{array}$$

Hence, **A = 5, B = 0 and C = 1.**

Question 8(i)

In each of the following cases, find the least value/values of letters used in place of digits :

$$\begin{array}{r} A B \\ \times 5 \\ \hline C A B \end{array}$$

Answer

$$\begin{array}{r} A B \\ \times 5 \\ \hline C A B \end{array}$$

Firstly, find the value of B.

$$B \times 5 = B$$

$$0 \times 5 = 0$$

$$5 \times 5 = 25$$

Possible values of B = 0 and 5.

Now, find the value of A

Lets take B = 0

$$\begin{array}{r} A \quad 0 \\ \times \quad 5 \\ \hline C \quad A \quad 0 \end{array}$$

$$A \times 5 = CA$$

$$5 \times 5 = 25$$

Hence, A = 5 and C = 2

Now lets take B = 5

$$\begin{array}{r} \overset{2}{A} \quad 5 \\ \times \quad 5 \\ \hline C \quad A \quad 5 \end{array}$$

$$A \times 5 + 2 = CA$$

No value for A satisfies this equation. Hence, B $\neq$ 5

$$\begin{array}{r}
 50 \\
 \times 5 \\
 \hline
 250
 \end{array}$$

A = 5, B = 0 and C = 2

Question 8(j)

In each of the following cases, find the least value/values of letters used in place of digits :

$$\begin{array}{r}
 8A5 \\
 + 94A \\
 \hline
 1A33
 \end{array}$$

Answer

$$\begin{array}{r}
 8A5 \\
 + 94A \\
 \hline
 1A33
 \end{array}$$

Clearly, we will find the value of A.

5 + A is a number whose ones digit is 3.

$\Rightarrow 5 + A = 3, 5 + A = 13, 5 + A = 23$; and so on.

$\Rightarrow A = 3 - 5, A = 13 - 5, A = 23 - 5$; and so on.

$\Rightarrow A = -2, A = 8, A = 18$; and so on.

$$\begin{array}{r}
 885 \\
 + 948 \\
 \hline
 1833
 \end{array}$$

Since, A is a digit. $\therefore A = 8$

Question 8(k)

In each of the following cases, find the least value/values of letters used in place of digits :

$$\begin{array}{r} 6 A B 5 \\ + D 5 8 C \\ \hline 9 3 5 1 \end{array}$$

Answer

$$\begin{array}{r} 6 A B 5 \\ + D 5 8 C \\ \hline 9 3 5 1 \end{array}$$

Firstly, we will find the value of C.

$5 + C$ is a number whose ones digit is 1.

$\Rightarrow 5 + C = 1, 5 + C = 11, 5 + C = 21$; and so on.

$\Rightarrow C = 1 - 5, C = 11 - 5, C = 21 - 5$; and so on.

$\Rightarrow C = -4, C = 6, C = 16$; and so on.

Since, C is a digit. $\therefore C = 6$

$$\begin{array}{r} 6 A B^1 5 \\ + D 5 8 6 \\ \hline 9 3 5 1 \end{array}$$

Secondly, we will find the value of B.

$B + 8 + 1$ is a number whose ones digit is 5.

$\Rightarrow B + 9 = 5$, $B + 9 = 15$, $B + 9 = 25$; and so on.

$\Rightarrow B = 5 - 9$, $B = 15 - 9$, $B = 25 - 9$; and so on.

$\Rightarrow B = -4$, $B = 6$, $B = 16$; and so on.

Since, B is a digit. $\therefore B = 6$

$$\begin{array}{r} \overset{1}{6} \overset{1}{A} \overset{1}{6} 5 \\ + D 5 8 6 \\ \hline 9 3 5 1 \end{array}$$

Thirdly, we will find the value of A.

$A + 5 + 1$ is a number whose ones digit is 3.

$\Rightarrow A + 6 = 3$, $A + 6 = 13$, $A + 6 = 23$; and so on.

$\Rightarrow A = 3 - 6$, $A = 13 - 6$, $A = 23 - 6$; and so on.

$\Rightarrow A = -3$, $A = 7$, $A = 17$; and so on.

Since, A is a digit. $\therefore A = 7$

$$\begin{array}{r} \overset{1}{6} \overset{1}{7} \overset{1}{6} 5 \\ + D 5 8 6 \\ \hline 9 3 5 1 \end{array}$$

Lastly, we will find the value of D.

$6 + D + 1$ is a number whose ones digit is 9.

$\Rightarrow D + 7 = 9$, $D + 7 = 19$, $D + 7 = 29$; and so on.

$\Rightarrow D = 9 - 7$, $D = 19 - 7$, $D = 29 - 7$; and so on.

$\Rightarrow D = 2$, $D = 12$, $D = 22$; and so on.

Since, D is a digit. $\therefore D = 2$

$$\begin{array}{r} \overset{1}{6} \quad \overset{1}{7} \quad \overset{1}{6} \quad 5 \\ + \quad 2 \quad 5 \quad 8 \quad 6 \\ \hline 9 \quad 3 \quad 5 \quad 1 \end{array}$$

A = 7, B = 6, C = 6 and D = 2.

Question 9

For a 3-digit number abc , what will be the quotient if $abc - cba$ is divided by 11 ?

Answer

As we know $abc = 100 \times a + 10 \times b + c$.

Similarly, $cba = 100 \times c + 10 \times b + a$.

Hence,

$$\begin{aligned} & \frac{abc - cba}{11} \\ &= \frac{(100 \times a + 10 \times b + c) - (100 \times c + 10 \times b + a)}{11} \\ &= \frac{100a + 10b + c - 100c - 10b - a}{11} \\ &= \frac{(100a - a) + (10b - 10b) + (c - 100c)}{11} \end{aligned}$$

$$= \frac{(99a) + (-99c)}{11}$$

$$= \frac{99(a - c)}{11}$$

$$= 9(a - c)$$

Hence, if $abc - cba$ is divided by 11 then the quotient is $9(a-c)$.

Exercise 5(B)

Question 1(i)

Which of the following number(s) are divisible by 2 :

1. 192

2. 1660

3. 1101

Answer

According to the divisibility of 2, if the unit digit of number is zero or an even number then the number is divisible by 2.

1. 192 Last digit is 2 (even number).
2. 1660 Last digit is 0.
3. 1101 Last digit is 1 (odd number)

Hence, 192 and 1101 are divisible by 2.

Question 1(ii)

Which of the following number(s) are divisible by 3 :

1. 261
2. 777
3. 6657

Answer

According to the divisibility of 3, if the sum of the digits of the number is divisible by 3 then the number is divisible by 3.

1. $261 = 2 + 6 + 1 = 9$. And 9 is divisible by 3
2. $777 = 7 + 7 + 7 = 21$. And 21 is divisible by 3
3. $6657 = 6 + 6 + 5 + 7 = 24$. And 24 is divisible by 3

Hence, all the three numbers, 261, 777 and 6657 are divisible by 3.

Question 1(iii)

Which of the following number(s) are divisible by 6 :

1. 2292

2. 2612

3. 5940

Answer

According to the divisibility of 6, if the number is divisible by 2 and 3 both then the number is divisible by 6.

1. 2292

Last digit is 2, hence 2292 is divisible by 2

$2 + 2 + 9 + 2 = 15$, 15 is divisible by 3. So, 2292 is divisible by 3

Hence, 2292 is divisible by 6.

2. 2612

Last digit is 2, hence 2612 is divisible by 2

$2 + 6 + 1 + 2 = 11$, 11 is not divisible by 3. So, 2612 is not divisible by 3

Hence, 2612 is not divisible by 6.

3. 5940

Last digit is 0, hence 5940 is divisible by 2

$5 + 9 + 4 + 0 = 18$, 18 is divisible by 3. So, 5940 is divisible by 3

Hence, 5940 is divisible by 6.

Hence, option 1 and 3 are the correct option.

Question 1(iv)

Which of the following number(s) are divisible by 9 :

1. 1863
2. 2064
3. 6111

Answer

According to the divisibility of 9, if the sum of the digits of the number is divisible by 9 then the number is divisible by 9.

1. $1863 = 1 + 8 + 6 + 3 = 18$. And 18 is divisible by 9.
2. $2064 = 2 + 0 + 6 + 4 = 12$. And 12 is not divisible by 9.
3. $6111 = 6 + 1 + 1 + 1 = 9$. And 9 is divisible by 9.

Hence, 1863 and 6111 are divisible by 9.

Question 1(v)

Which of the following number(s) are divisible by 4 :

1. 360
2. 4093
3. 5348

Answer

According to the divisibility of 4, if the two digit number formed by its ten's digit and unit digit is divisible by 4 then the number is divisible by 4.

1. 360 — Last 2 digits are 60 and 60 is divisible by 4.
2. 4093 — Last 2 digits are 93 and 93 is not divisible by 4.
3. 5348 — Last 2 digits are 48 and 48 is divisible by 4.

Hence, 360 and 5348 are divisible by 4.

Question 1(vi)

Which of the following number(s) are divisible by 5 :

1. 3250
2. 5557
3. 39255

Answer

According to the divisibility of 5, if the unit digit is 0 or 5 then the number is divisible by 5.

1. 3250 — Last digit is 0.
2. 5557 — Last digit is 7.
3. 39255 — Last digit is 5.

Hence, 3250 and 39255 are divisible by 5.

Question 1(vii)

Which of the following number(s) are divisible by 10 :

1. 5100

2. 4612

3. 3400

Answer

According to the divisibility of 10, if the unit digit is 0 then the number is divisible by 10.

1. 5100 — Last digit is 0.

2. 4612 — Last digit is 2.

3. 3400 — Last digit is 0.

Hence, 5100 and 3400 are divisible by 10.

Question 1(viii)

Which of the following number(s) are divisible by 11 :

1. 2563

2. 8307

3. 95635

Answer

According to the divisibility of 11, if the difference between the sum of its digits in even places and the sum of its digits in odd places is either 0 or divisible by

11 then the number is divisible by 11.

1. 2563 — the sum of its digits in odd places = $2 + 6 = 8$

the sum of its digits in even places = $5 + 3 = 8$

The difference = $8 - 8 = 0$.

2. 8307 — the sum of its digits in odd places = $8 + 0 = 8$

the sum of its digits in even places = $3 + 7 = 10$

The difference = $10 - 8 = 2$ (Not divisible by 11)

3. 95635 — the sum of its digits in odd places = $9 + 6 + 5 = 20$

the sum of its digits in even places = $5 + 3 = 8$

The difference = $20 - 8 = 12$ (Not divisible by 11)

Hence, 2563 is divisible by 11.

Question 2

Show that the number 21952 is divisible by 7.

Answer

According to the divisibility of 7, take the last digit of the number, double it then subtract the result from the rest of the number, if the resulting number is divisible by 7 then the number is divisible by 7.

The number is 21952.

The last digit is 2 and its double is 4.

Rest of the number = 2195

Subtract 4 from 2195 to get $2195 - 4 = 2191$.

Take 2191 as the number.

The last digit is 1 and its double is 2.

Rest of the number = 219

Subtract 2 from 219 to get $219 - 2 = 217$

Take 217 as the number.

The last digit is 7 and its double is 14.

Rest of the number = 21

Subtract 14 from 21 to get $21 - 14 = 7$ and 7 is divisible by 7.

Hence, 21952 is divisible by 7.

Question 3

Show that the number 6486 is not divisible by 7.

Answer

According to the divisibility of 7, take the last digit of the number, double it then subtract the result from the rest of the number, if the resulting number is divisible by 7 then the number is divisible by 7.

The number is 6486.

The last digit is 6 and its double is 12.

Rest of the number = 648

Subtract 12 from 648 to get $648 - 12 = 636$.

Take 636 as the number.

The last digit is 6 and its double is 12.

Rest of the number = 63

Subtract 12 from 63 to get $63 - 12 = 51$.

51 is not divisible by 7.

Hence, 6486 is not divisible by 7.

Question 4(i)

For what value of digit x is :

$1x5$ divisible by 3 ?

Answer

According to the divisibility of 3, if the sum of the digits of the number is divisible by 3 then the number is divisible by 3.

$1x5 = 1 + x + 5 = 6 + x$. And $(6 + x)$ should be a multiple of 3.

Hence, $6 + x = 0, 3, 6, 9, 12, 15, 18, \dots$

$x = -6, -3, 0, 3, 6, 9, 12, \dots$

Since, x is a digit. $\therefore x = 0, 3, 6 \text{ or } 9$

Question 4(ii)

For what value of digit x is :

$31x5$ divisible by 3 ?

Answer

According to the divisibility of 3, if the sum of the digits of the number is divisible by 3 then the number is divisible by 3.

$31x5 = 3 + 1 + x + 5 = 9 + x$. And $(9 + x)$ should be a multiple of 3.

Hence, $9 + x = 0, 3, 6, 9, 12, 15, 18, \dots$

$x = -9, -6, -3, 0, 3, 6, 9, \dots$

Since, x is a digit. $\therefore x = 0, 3, 6 \text{ or } 9$

Question 4(iii)

For what value of digit x is :

$28x6$ a multiple of 3 ?

Answer

According to the divisibility of 3, if the sum of the digits of the number is divisible by 3 then the number is divisible by 3.

$28x6 = 2 + 8 + x + 6 = 16 + x$. And $(16 + x)$ should be a multiple of 3.

Hence, $16 + x = 0, 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, \dots$

$x = -16, -13, -10, -7, -4, -1, 2, 5, 8, 11, \dots$

Since, x is a digit. $\therefore x = 2, 5 \text{ or } 8$

Question 4(iv)

For what value of digit x is :

24x divisible by 6 ?

Answer

According to the divisibility of 6, if the number is divisible by 2 and 3 both then the number is divisible by 6.

24x is a multiple of 2 and 3 both.

To make 24x divisible by 2 $\Rightarrow$ x should be 0 or an even number.

So, $x = 0, 2, 4, 6$ or $8 \dots\dots\dots(1)$

To make 24x divisible by 3 $\Rightarrow 2 + 4 + x$ is a multiple of 3.

$\Rightarrow 6 + x$ is a multiple of 3.

$\Rightarrow 6 + x = 0, 3, 6, 9, 12, 15, 18, \dots\dots\dots$

$\Rightarrow x = -6, -3, 0, 3, 6, 9, 12, \dots\dots\dots$

Since x is a digit. So, $x = 0, 3, 6, 9 \dots\dots\dots(2)$

From (1) and (2), **$x = 0$ or 6 .**

Question 4(v)

For what value of digit x is :

3x26 a multiple of 6 ?

Answer

According to the divisibility of 6, if the number is divisible by 2 and 3 both then the number is divisible by 6.

$3x26$ should be a multiple of 2 and 3 both.

$3x26$ is divisible by 2 as it has even number 6 at its unit place.

To make $3x26$ divisible by 3 $\Rightarrow 3 + x + 2 + 6$ should be a multiple of 3.

$\Rightarrow 11 + x$ should be a multiple of 3.

$\Rightarrow 11 + x = 0, 3, 6, 9, 12, 15, 18, \dots$

$\Rightarrow x = -11, -8, -5, -2, 1, 4, 7, \dots$

Since x is a digit. **So, $x = 1, 4, \text{ or } 7$.**

Question 4(vi)

For what value of digit x is :

$42x8$ divisible by 4 ?

Answer

According to the divisibility of 4, if the two digit number formed by its ten's digit and unit digit is divisible by 4 then the number is divisible by 4.

$42x8$ is divisible by 4

$\Rightarrow x8$ is divisible by 4

$\Rightarrow 10x + 8$ is divisible by 4

$\Rightarrow 10x + 8 = 8, 28, 48, 68, 88, 108, \dots$

$\Rightarrow 10x = 0, 20, 40, 60, 80, 100, \dots$

$\Rightarrow x = 0, 2, 4, 6, 8, 10, \dots$

Since x is a digit. **So, $x = 0, 2, 4, 6$ or 8 .**

Question 4(vii)

For what value of digit x is :

9142x a multiple of 4 ?

Answer

According to the divisibility of 4, if the two digit number formed by its ten's digit and unit digit is divisible by 4 then the number is divisible by 4.

9142x is divisible by 4

$\Rightarrow 2x$ is divisible by 4

$\Rightarrow 20 + x$ is divisible by 4

$\Rightarrow 20 + x = 20, 24, 28, 32, 36, 40, \dots$

$\Rightarrow x = 0, 4, 8, 12, 16, \dots$

Since x is a digit. **So, $x = 0, 4$ or 8 .**

Question 4(viii)

For what value of digit x is :

7x34 divisible by 9 ?

Answer

According to the divisibility of 9, if the sum of the digits of the number is divisible by 9 then the number is divisible by 9.

$\therefore 7x34$ is divisible by 9

$\Rightarrow 7 + x + 3 + 4$ is divisible by 9

$\Rightarrow 14 + x$ is divisible by 9

$\Rightarrow 14 + x = 0, 9, 18, 27, 36, \dots$

$\Rightarrow x = -14, -5, 4, 13, 22, \dots$

Since x is a digit. **So, $x = 4$.**

Question 4(ix)

For what value of digit x is :

$5x555$ a multiple of 9 ?

Answer

According to the divisibility of 9, if the sum of the digits of the number is divisible by 9 then the number is divisible by 9.

$\therefore 5x555$ is divisible by 9

$\Rightarrow 5 + x + 5 + 5 + 5$ is divisible by 9

$\Rightarrow 20 + x$ is divisible by 9

$\Rightarrow 20 + x = 0, 9, 18, 27, 36, \dots$

$\Rightarrow x = -20, -11, -2, 7, 16, \dots$

Since x is a digit. **So, $x = 7$.**

Question 4(x)

For what value of digit x is :

$3x2$ divisible by 11 ?

Answer

According to the divisibility of 11, if the difference between the sum of its digits in even places and the sum of its digits in odd places is either 0 or divisible by 11 then the number is divisible by 11.

$\therefore 3x2$ is divisible by 11

$\Rightarrow$ Sum of the digits in even place = x

Sum of the digits in odd place = $3 + 2 = 5$

Difference between the sum of the digits in odd places and the sum of the digits in even places = $5 - x$

$\Rightarrow 5 - x$ must be divisible by 11

$\Rightarrow 5 - x = 0, 11, 22, 33, \dots$

$\Rightarrow x = 5, -6, -17, -28, \dots$

Since x is a digit. **So, $x = 5$.**

Question 4(xi)

For what value of digit x is :

$5x2$ a multiple of 11 ?

Answer

According to the divisibility of 11, if the difference between the sum of its digits in even places and the sum of its digits in odd places is either 0 or divisible by 11 then the number is divisible by 11.

$\therefore 5x2$ is divisible by 11

$\Rightarrow$ Sum of the digits in even place = x

Sum of the digits in odd place = $5 + 2 = 7$

Difference between the sum of the digits in odd places and the sum of the digits in even places = $7 - x$

$\Rightarrow 7 - x$ must be divisible by 11

$\Rightarrow 7 - x = 0, 11, 22, 33, \dots$

$\Rightarrow x = 7, -4, -15, -26, \dots$

Since x is a digit. **So, $x = 7$.**

Test Yourself

Question 1(i)

ab and ba are two 2-digit numbers then $ab + ba$ is equal to :

1. $2ab$
2. $11ab$
3. $11(a + b)$
4. none of these

Answer

$$ab + ba$$

$$= (10 \times a + b) + (10 \times b + a)$$

$$= 10a + b + 10b + a$$

$$= (10a + a) + (b + 10b)$$

$$= 11a + 11b$$

$$= 11(a + b)$$

Hence, option 3 is the correct option.

Question 1(ii)

In a 2-digit number ab and $a > b$ then $ab - ba$ is equal to :

1. 0
2. $9(a - b)$

3. $9(b - a)$

4. none of these

Answer

$$ab - ba$$

$$= (10a + b) - (10b + a)$$

$$= 10a + b - 10b - a$$

$$= (10a - a) + (b - 10 \times b)$$

$$= 9a - 9 \times b$$

$$= 9(a - b)$$

Hence, option 2 is the correct option.**Question 1(iii)**If $a = b$, then $ab \div ba$ is equal to :

1. 0

2. $9(a - b)$

3. $11(a + b)$

4. none of these

Answer

$$ab \div ba$$

$$= \frac{ab}{ba}$$

$$= \frac{a \times b}{b \times a}$$

$$= \frac{a \times a}{a \times a}$$

$$= \frac{a^2}{a^2}$$

$$= \frac{\cancel{a^2}}{\cancel{a^2}}$$

= 1 Hence, option 4 is the correct option.

Question 1(iv)

If $a > b$, then for two 3-digit numbers abc and cba ; we have $abc - cba$ equal to :

1. -27

2. -6

3. 6

4. none of these

Answer

$$abc - cba$$

$$= (100a + 10b + c) - (100c + 10b + a)$$

$\Rightarrow A = 8, A = 18, A = 28$; and so on

Since A is a digit. So $A = 8$

Hence, option 3 is the correct option.

Question 1(vi)

Without actual calculation, the quotient, when the sum of 59 and 95 is divided by 11, is :

1. 14
2. 9
3. $95 - 59$
4. none of these

Answer

According to the property, when $ab + ba$ is divided by $a + b$, the quotient = 11.

So, when $59 + 95$ is divided by 11, the quotient = $(5 + 9) = 14$.

Hence, option 1 is the correct option.

Question 1(vii)

Without actual calculation, the quotient when $74 - 47$ is divided by 9, is :

1. 11
2. 7
3. 3

4. none of these

Answer

According to the property, when $ab - ba$ is divided by 9, the quotient = $a - b$.

So, when $74 - 47$ is divided by 9, the quotient = $7 - 4 = 3$.

Hence, option 3 is the correct option.

Question 1(viii)

The smallest two digit number $10a + 4$ is divisible by 6, if $a + 4$ is divisible by :

1. 2

2. 6

3. 3

4. none of these

Answer

According to the divisibility of 6, if the number is divisible by 2 and 3 both then the number is divisible by 6.

$10a + 4$ is divisible by 2 as 4 (even number) is at its unit place.

$10a + 4$ is divisible by 3 when $a + 4$ is divisible by 3.

Then, $a + 4$ is divisible by 6.

Hence, option 2 is the correct option.

Question 1(ix)

The number $3z4$ is divisible by 9, if the digit z is equal to :

1. 7
2. 2
3. 9
4. none of these

Answer

According to the divisibility of 9, if the sum of the digits of the number is divisible by 9 then the number is divisible by 9.

$\therefore 3z4$ is divisible by 9

$\Rightarrow 3 + z + 4$ is divisible by 9

$\Rightarrow 7 + z$ is divisible by 9

$\Rightarrow 7 + z = 0, 9, 18, 27, 36, \dots$

$\Rightarrow z = -7, 2, 11, 20, \dots$

Since z is a digit. So, $z = 2$.

Hence, option 2 is the correct option.

Question 1(x)

Statement 1: When the sum of a two-digit number and the number obtained by reversing its digit is divided by 11, the quotient is equal to the sum of two digits.

Statement 2: When the sum of a two digit number and the number obtained by reversing its digits is divided by the sum of the two digits, the quotient is always 11.

Which of the following options is correct ?

1. Both the statements are true.
2. Both the statements are false.
3. Statement 1 is true, and statement 2 is false.
4. Statement 1 is false, and statement 2 is true.

Answer

Let the digit at ten's place be x and at unit's place be y .

The two-digit number = $10 \times x + y = 10x + y$

Number obtained by reversing its digits = $10 \times y + x = 10y + x$

Sum of the number and reversed numbers = $(10x + y) + (10y + x)$

$$= 10x + x + y + 10y$$

$$= 11x + 11y$$

$$= 11(x + y)$$

On dividing by 11, we get :

$$\frac{11(x + y)}{11} = (x + y)$$

The quotient is equal to the sum of the digits.

So, statement 1 is true.

On dividing the sum of numbers by sum of the digits = $\frac{11(x + y)}{x + y} = 11$

The quotient is equal to 11.

So, statement 2 is true.

Hence, option 1 is the correct option.

Question 1(xi)

Assertion (A) : $759 = 100 \times 7 + 10 \times 5 + 1 \times 9$.

Reason (R) : In a three-digit number $100p + 10q + 1r$, the digit p at hundred's place is any whole number from 1 to 9, the digit q at ten's place is any whole number from 0 to 9 and the digit r at unit place is any whole number from 0 to 9.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Given,

$$759 = 100 \times 7 + 10 \times 5 + 1 \times 9$$

Solving R.H.S. of the above equation, we get :

$$\Rightarrow 100 \times 7 + 10 \times 5 + 1 \times 9$$

$$\Rightarrow 700 + 50 + 9$$

$$\Rightarrow 759$$

Since, L.H.S. = R.H.S.

So, assertion (A) is true.

In a three-digit number $100p + 10q + 1r$,

q and r can be 0 but p cannot be equal to 0 for 3-digit number.

So, reason (R) is false.

Hence, option 3 is the correct option.

Question 1(xii)

Assertion (A) : If $36p52q9$ is divisible by 9, then $p + q = 2$.

Reason (R) : A number is divisible by 3 if the sum of its digits is divisible by 3.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Divisibility Rule for 9 : A number is divisible by 9 if the sum of its digits is divisible by 9.

So, if $36p52q9$ is divisible by 9, then the sum of digits of $36p52q9$, will also be divisible by 9.

$$\Rightarrow 3 + 6 + p + 5 + 2 + q + 9$$

$$\Rightarrow 25 + p + q$$

$$\Rightarrow 27 \text{ (If, } p + q = 2 \text{)}$$

Since, 27 is divisible by 3, so if $36p52q9$ is divisible by 9, then $p + q = 2$.

So, assertion (A) is true.

We know that,

A number is divisible by 3 if the sum of its digits is divisible by 3.

This is a correct rule for divisibility by 3.

So, reason (R) is true, but it does not explain assertion.

Hence, option 2 is the correct option.

Question 1(xiii)

Assertion (A) : Factors of the sum of a three-digit number 542 and the numbers obtained by changing the order of the digits cyclically are 1, 11, 111, $5 + 4 + 2$.

Reason (R) : The sum of a three-digit number and the two numbers obtained by changing the digit cyclically is completely divisible by (i) 11, (ii) 111 and (iii) the sum of the digits.

1. Both A and R are correct, and R is the correct explanation for A.

2. Both A and R are correct, and R is not the correct explanation for A.

3. A is true, but R is false.

4. A is false, but R is true.

Answer

Given a three-digit number : 542

The cyclic permutations of the digits are : 542, 425 and 254.

Now we compute the sum of these three numbers : $542 + 425 + 254 = 1221$

$$1221 = 1 \times 11 \times 111$$

So, assertion (A) is true.

Let the given a three-digit number : $100p + 10q + r$

The cyclic permutations of the digits are: $100p + 10q + r$, $100q + 10r + p$ and $100r + 10p + q$

Now we compute the sum of these three numbers: $(100p + 10q + r) + (100q + 10r + p) + (100r + 10p + q)$

$$= (100p + 10p + p) + (100q + 10q + q) + (100r + 10r + r)$$

$$= 111p + 111q + 111r$$

$$= 111(p + q + r)$$

$$\text{Factors of } 111(p + q + r) = 1 \times 111 \times (p + q + r)$$

According to reason (R), 111 and sum of the digits are the factors of the sum of a three - digit number and the two number obtained by changing the digit cyclically, but 11 is not.

So, reason (R) is false.

Hence, option 3 is the correct option.

Question 1(xiv)

Assertion (A) : 2574 is divisible by 11 but 7083 is not divisible by 11.

Reason (R) : A number is divisible by 11 if the difference between the sum of its digits in even places and the sum of its digits in odd places is either 0 or divisible by 11.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

We know that,

A number is divisible by 11 if the difference between the sum of its digits in even places and the sum of its digits in odd places is either zero or divisible by 11.

So, reason (R) is true.

Lets check 2574 is divisible by 11. Digits at even place (5 and 4), digits at odd place (2 and 7).

$$\Rightarrow (5 + 4) - (2 + 7)$$

$$\Rightarrow 9 - 9$$

$$\Rightarrow 0.$$

Since, the difference is 0,

$\therefore$ 2574 is divisible by 11.

Lets check 7083 is divisible by 11. Digits at even place (0 and 3), digits at odd place (7 and 8).

$$\Rightarrow (0 + 3) - (7 + 8)$$

$$\Rightarrow 3 - 15$$

$$\Rightarrow -12$$

Since, difference is neither 0 nor divisible by 11,

$\therefore$ 7083 is not divisible by 11.

So, assertion (A) is true and reason explain assertion.

Hence, option 1 is the correct option.

Question 2

For three 3-digit numbers abc, cab and bca, show that $abc + cab + bca$ is divisible by 37.

Answer

$$abc + cab + bca$$

$$= (100a + 10b + c) + (100c + 10a + b) + (100b + 10c + a)$$

$$= 100a + 10b + c + 100c + 10a + b + 100b + 10c + a$$

$$= (100a + 10a + a) + (10b + b + 100b) + (c + 100c + 10c)$$

$$= 111a + 111b + 111c$$

$$= 111(a + b + c)$$

$$= 3 \times 37 \times (a + b + c)$$

Hence, $(abc + cab + bca)$ is divisible by 37

Question 3

If the three digit number $24x$ is divisible by 9, find the value of digit x .

Answer

According to the divisibility of 9, if the sum of the digits of the number is divisible by 9 then the number is divisible by 9.

$\therefore 24x$ is divisible by 9

$\Rightarrow 2 + 4 + x$ is divisible by 9

$\Rightarrow 6 + x$ is divisible by 9

$\Rightarrow 6 + x = 0, 9, 18, 27, 36, \dots$

$\Rightarrow x = -6, 3, 12, 21, 30, \dots$

Since x is a digit. **So, $x = 3$.**

Question 4

If $31y5$ is divisible by 3, find the least value of digit y .

Answer

According to the divisibility of 3, if the sum of the digits of the number is divisible by 3 then the number is divisible by 3.

$\therefore 31y5$ is divisible by 3

$\Rightarrow 3 + 1 + y + 5$ is divisible by 3

$\Rightarrow 9 + y$ is divisible by 3

$\Rightarrow 9 + y = 0, 3, 6, 9, 12, \dots\dots\dots$

$\Rightarrow y = -9, -6, -3, 0, 3, 6, 9, \dots\dots\dots$

Since y is a digit, so, $y = 0, 3, 6$ or 9 .

Hence, least value of digit $y = 0$.

Question 5

In a 3-digit number, the ten's digit is thrice the unit digit and hundred's digit is twice the unit digit. If the sum of all the three digits is 12, find the number.

Answer

Let the 3-digit number be abc .

$$abc = 100a + 10b + c$$

Given,

$$b = 3c \dots\dots\dots(1)$$

$$a = 2c \dots\dots\dots(2)$$

and

$$a + b + c = 12 \text{(3)}$$

Putting value of b and a from eqns (1) and (2) into eq (3) we get,

$$2c + 3c + c = 12$$

$$\Rightarrow 6c = 12$$

$$\Rightarrow c = \frac{12}{6}$$

$$\Rightarrow c = 2$$

Putting value of c in eq 1,

$$b = 3c = 3 \times 2 = 6$$

Putting value of c in eq 2,

$$a = 2c = 2 \times 2 = 4$$

Hence, the number is 462.

Question 6

Find the digits A, B and C, if

$$\begin{array}{r} 7 \quad 3 \quad A \\ - \quad B \quad C \quad 9 \\ \hline 3 \quad 4 \quad 8 \end{array}$$

Answer

$$\begin{array}{r} 7 \quad 3 \quad A \\ - \quad B \quad C \quad 9 \\ \hline 3 \quad 4 \quad 8 \end{array}$$

Firstly, we will find the value of A.

Clearly, $A - 9$ is a number whose ones digit is 8.

$$\Rightarrow A - 9 = 8, A - 9 = 18, A - 9 = 28; \text{ and so on.}$$

$$\Rightarrow A = 8 + 9, A = 18 + 9, A = 28 + 9; \text{ and so on.}$$

$$\Rightarrow A = 17, A = 27, A = 37; \text{ and so on.}$$

Since, A is a digit. $\therefore A = 7$.

Secondly, we will find the value of C.

$$\begin{array}{r} 7 ^2\cancel{3} ^{17}\cancel{7} \\ - B C 9 \\ \hline 3 4 8 \end{array}$$

Clearly, $2 - C$ is a number whose ones digit is 4.

As it is not possible to subtract any number from 2 that gives 4. Hence, borrowing one number from preceding digit gives $12 - C$.

$$\begin{array}{r} ^6\cancel{7} ^{12}\cancel{3} ^{17}\cancel{7} \\ - B C 9 \\ \hline 3 4 8 \end{array}$$

$$\Rightarrow 12 - C = 4, 12 - C = 14, 12 - C = 24; \text{ and so on.}$$

$$\Rightarrow C = 12 - 4, C = 12 - 14, C = 12 - 24; \text{ and so on.}$$

$$\Rightarrow C = 8, C = -2, C = -12; \text{ and so on.}$$

Since, C is a digit. $\therefore C = 8$.

Lastly, we will find the value of B.

Clearly, $6 - B$ is a number whose ones digit is 3.

$$\Rightarrow 6 - B = 3, 6 - B = 13, 6 - B = 23; \text{ and so on.}$$

$$\Rightarrow B = 6 - 3, B = 6 - 13, B = 6 - 23; \text{ and so on.}$$

$$\Rightarrow B = 3, B = -7, B = -17; \text{ and so on.}$$

Since, B is a digit. $\therefore B = 3$.

$$\begin{array}{r} 7 \quad 3 \quad 7 \\ - \quad 3 \quad 8 \quad 9 \\ \hline 3 \quad 4 \quad 8 \end{array}$$

Hence, **A = 7, B = 3 and C = 8.**

Question 7(i)

Find the digits A and B, if :

$$\begin{array}{r} 2 \quad A \quad B \\ + \quad A \quad B \quad 1 \\ \hline B \quad 1 \quad 8 \end{array}$$

Answer

$$\begin{array}{r} 2 \quad A \quad B \\ + \quad A \quad B \quad 1 \\ \hline B \quad 1 \quad 8 \end{array}$$

Firstly, we will find the value of B

Clearly, $B + 1$ is a number whose ones digit is 8.

$$\Rightarrow B + 1 = 8, B + 1 = 18, B + 1 = 28; \text{ and so on.}$$

$$\Rightarrow B = 8 - 1, B = 18 - 1, B = 28 - 1; \text{ and so on.}$$

$$\Rightarrow B = 7, B = 17, B = 27; \text{ and so on.}$$

Since, B is a digit. $\therefore B = 7$

$$\begin{array}{r} 2 \quad A \quad 7 \\ + \quad A \quad 7 \quad 1 \\ \hline 7 \quad 1 \quad 8 \end{array}$$

Now, we find the value of A.

Clearly, $A + 7$ is a number whose ones digit is 1.

$$\Rightarrow A + 7 = 1, A + 7 = 11, A + 7 = 21; \text{ and so on.}$$

$$\Rightarrow A = 1 - 7, A = 11 - 7, A = 21 - 7; \text{ and so on.}$$

$$\Rightarrow A = -6, A = 4, A = 14; \text{ and so on.}$$

Since, A is a digit. $\therefore A = 4$

$$\begin{array}{r} 2 \quad 4 \quad 7 \\ + \quad 4 \quad 7 \quad 1 \\ \hline 7 \quad 1 \quad 8 \end{array}$$

Hence, $A = 4$ and $B = 7$.

Question 7(ii)

Find the digits A and B, if :

$$\begin{array}{r}
 \text{A} \quad \text{B} \\
 + \quad 3 \quad 7 \\
 \hline
 9 \quad \text{A} \\
 \hline
 \end{array}$$

Answer

$$\begin{array}{r}
 \text{A} \quad \text{B} \\
 + \quad 3 \quad 7 \\
 \hline
 9 \quad \text{A} \\
 \hline
 \end{array}$$

Lets take A as a 1-digit number

$$B + 7 = A \dots\dots(1)$$

$$A + 3 = 9 \dots\dots(2)$$

$$A = 9 - 3 = 6$$

Putting the value of A in equation (1), we get

$$B + 7 = 6$$

$$B = 6 - 7$$

$$B = -1$$

$\therefore$ As we know B is a digit. Hence $B \neq -1$.

Lets take A as a 2-digit number with ones digit as 1.

We can represent A as $10 + A$

$$\begin{array}{r}
 \overset{1}{\text{A}} \quad \text{B} \\
 + \quad 3 \quad 7 \\
 \hline
 9 \quad \text{A} \\
 \hline
 \end{array}$$

$$B + 7 = 10 + A \dots\dots(3)$$

$$1 + A + 3 = 9 \text{ (one carry on ten's digit) } \dots\dots(4)$$

$$A + 4 = 9$$

$$A = 9 - 4 = 5$$

Putting the value of A in equation 3.

$$B + 7 = 10 + 5$$

$$B + 7 = 15$$

$$B = 15 - 7 = 8$$

Hence, $A = 5$, $B = 8$

Now, lets take A as a 2-digit number with ones digit as 2.

We can represent A as $20 + A$

$$\begin{array}{r} \overset{2}{A} \quad B \\ + \quad 3 \quad 7 \\ \hline 9 \quad A \end{array}$$

$$B + 7 = 20 + A \dots\dots(5)$$

$$2 + A + 3 = 9 \text{ (two carry on ten's digit) } \dots\dots(6)$$

$$A + 5 = 9$$

$$A = 9 - 5 = 4$$

Putting the value of A in equation (5),

$$B + 7 = 20 + 4$$

$$B + 7 = 24$$

$$B = 24 - 7 = 17$$

$\therefore$ As we know B is a digit. Hence $B \neq 17$.

$$\begin{array}{r} 5 \ 8 \\ + \ 3 \ 7 \\ \hline 9 \ 5 \end{array}$$

Hence, **A = 5 , B = 8.**

Question 7(iii)

Find the digits A and B, if :

$$\begin{array}{r} 1 \ B \\ \times \ B \\ \hline 9 \ B \end{array}$$

Answer

$$\begin{array}{r} 1 \ B \\ \times \ B \\ \hline 9 \ B \end{array}$$

Clearly, $B \times B = B$

$$1 \times 1 = 1$$

$$5 \times 5 = 25$$

$$6 \times 6 = 36$$

So, possible answer for B = 1 or 5 or 6.

But 1 and 5 did not satisfy the multiplication.

$$\begin{array}{r} \overset{3}{1} \ 6 \\ \times \ 6 \\ \hline 9 \ 6 \end{array} \text{ Hence, } \mathbf{B = 6}.$$

Question 7(iv)

Find the digits A and B, if :

$$\begin{array}{r} 8 \ B \ 4 \\ - 5 \ 1 \ B \\ \hline B \ 2 \ 1 \end{array}$$

Answer

$$\begin{array}{r} 8 \ B \ 4 \\ - 5 \ 1 \ B \\ \hline B \ 2 \ 1 \end{array}$$

Clearly, $4 - B$ is a number whose ones digit is 1.

$\Rightarrow 4 - B = 1, 4 - B = 11, 4 - B = 21$; and so on.

$\Rightarrow B = 4 - 1, B = 4 - 11, B = 4 - 21$; and so on.

$\Rightarrow B = 3, B = -7, B = -17$; and so on.

$$\begin{array}{r} 8 \ 3 \ 4 \\ - 5 \ 1 \ 3 \\ \hline 3 \ 2 \ 1 \end{array}$$

Since, B is a digit. $\therefore \mathbf{B = 3}$

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